Sherlynette Pérez Castro, Ph.D.
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EDUCATION

PhD Ecology. 2018. Department of Biology, San Diego State University-University of California, Davis, CA

MS Soil Sciences. 2010. Department of Crops and Agroenvironmental Sciences, University of Puerto Rico, Mayagüez, PR,

BA Environmental Sciences. 2006. Department of Environmental Sciences, University of Puerto Rico, Río Piedras, PR

PROFESIONAL EXPERIENCE

2025 - present Postdoctoral Research Associate
Institute of Marine Sciences, Morehead City, NC, Noble Lab

2023-2025 Postdoctoral Research Associate
University of Georgia, Athens, Thompson Lab

2018-2023 Postdoctoral Scientist
Marine Biological Laboratory, Woods Hole, MA, Ruff & Cardon Lab

2010-2013 Soil Scientist
USDA Natural Resources Conservation Service, FL

PUBLICATIONS

13. **Pérez Castro, S**; Morikwe, UC; Hill, MS; Lott, MEJ; Dempsey EE; Gunsch, C; Noble, RT; Landreville, KD; Jeffers-Francis, L. (2026). Kitchen-scale connectivity dominates bacterial and fungal communities across sink surfaces. *BMC Microbiology*. 10.1186/s12866-026-05609-0
12. García Arredondo, MG; **Pérez Castro, S**; Aguirre Mendoza, F; Mardis, G; Koch, C; Tfaily, MM; Cardon, ZG; Keiluweit, M. (2026). Nutrient stress reduces the extent but increases the Intensity of root-driven destabilization of mineral-associated organic matter. *Communications Earth & Environment*. 10.1038/s43247-026-04008-7
11. Calabrese, S; Tulip, SS; **Pérez Castro, S**; Thompson, A; Pett-Ridge, J; Nikolov, Z. (2026). Fluxes versus pools: Connecting temperature dependence and sensitivity of soil carbon dynamics across timescales. *Journal of Geophysical Research: Biogeosciences*. 10.1029/2026JG009818
10. Tulip, SS; **Pérez Castro, S**; Thompson, A; Pett-Ridge, Calabrese, S. (2026). Quantifying temperature dependence of Fe reduction in humid tropical soils: a Bayesian model-data integration. *EGUsphere*. 10.5194/egusphere-2026-2111
9. Bölscher, T; Cardon, ZG; García Arredondo, M; Grand, S; Griffen, G; Hestrin, R;…; **Pérez Castro, S**; Persson, P; Riley, WJ; Keiluweit, M. (2025). Vulnerability of mineral-organic associations in the rhizosphere. *Nature Communications*. 10.1038/s41467-025-61273-4
8. Barcellos, D; **Pérez Castro, S**; Cambell, A; Kimbrel JA; Blazewicz SJ; Wollard, J; Pett-Ridge, J; Thompson A. (2025) Duration of O_2 exposure determines dominance of Fe^{II} vs CH_4 production in tropical forest soils. *Environmental Science & Technology*. 10.1021/acs.est.4c12329
7. **Pérez Castro, S**; López Peredo, E; Mason, O; Vineis, J; Bowen JL; Mortazavi, B; Ganesh, A; Ruff, E; Paul, B; Giblin, AE; Cardon, ZG. (2023). Diversity at single nucleotide to pangenome

scales among sulfur cycling bacteria in salt marshes. *Applied and Environmental Microbiology*. 10.1128/aem.00988-23

6. **Pérez Castro, S**; Borton, MA; Regan, K; Hrabe de Angelis, I; Wrighton, KC.; Teske, AP.; Strous, M; Ruff, SE. (2021). Degradation of biological macromolecules supports uncultured microbial populations in Guaymas Basin hydrothermal sediments. *The ISME Journal*. 10.1038/s41396-021-01026-5
5. Lipson DA; Raab, T; **Pérez Castro S**; Powell, A. (2021). Organohalide respiring bacteria at the heart of anaerobic metabolism in Arctic wet tundra soils. *Applied and Environmental Microbiology*. 10.1128/AEM.01643-20
4. Bhatnagar, S; Cowley, ES; Kopf, SH; **Pérez Castro, S**; Kearney S; Dawson, SC; Hanselmann, K; Ruff, SE. (2020). Microbial community dynamics and coexistence in a sulfide-driven phototrophic bloom. *Environmental Microbiome*. 10.1186/s40793-019-0348-0
3. **Pérez Castro, S**; Esch, EH; Eviner, VT; Cleland, EE; Lipson, DA. (2019). Exotic herbaceous species interact with severe drought to alter soil N cycling in a semi-arid shrubland. *Geoderma*. 10.1016/j.geoderma.2019.114111
2. **Pérez Castro, S**; Cleland, EE; Wagner, R; Al Sawad, R; Lipson, DA. (2019). Soil microbial responses to drought and exotic plants shift carbon metabolism. *The ISME Journal*. 10.1038/s41396-019-0389-9
1. Puritty, CE; Esch, EH; **Pérez Castro, S**; Lipson, DA; Cleland, EE. (2019). Drought in Southern California coastal sage scrub reduces biomass of exotic species more than native species, but exotic growth recovers quickly when drought ends. *Plant Ecology*. 10.1007/s11258-019-00912-5

PRESENTATIONS & POSTERS

2025 International Conference on Microbiome Engineering, Antwerp, Belgium (*Poster presentation*)
Microbial Dark Matter Symposium, Laguna Beach, CA (*Poster presentation*)
2023 ESS PI Meeting, Bethesda, MD (*Oral presentation*)
2022 AGU Fall Meeting, Chicago (*Oral presentation*)
Ecological Society of America, Montréal, Canada (*Oral presentation*)
Stockbridge School of Agriculture Seminar, UMass Amherst (*Invited speaker*)
2021 AGU Fall Meeting, New Orleans, LA (*Poster presentation*)
Ecological Society of America, Virtual (*Oral presentation*)
2019 Applied and Environmental Microbiology Gordon Research Seminar, South Hadley, MA (*Oral presentation*)
Gordon Research Conference, South Hadley, MA (*Poster presentation*)
2016 AGU Fall Meeting, San Francisco, CA (*Poster presentation*)
2010 ASA, CSSA, and SSSA International Annual Meeting, Long Beach, CA (*Poster presentation*)

TEACHING

2026 Human impacts on estuarine processes. **Guest lecturer**. Institute of Marine Sciences
2025 Data visualization and statistical computing using R. **Instructor**. University of North Carolina
2021 Semester in Environmental Sciences. **Co-Instructor**. Marine Biological Laboratory
2020 Microbiomes Across Environments. **Teaching assistant**. Marine Biological Laboratory
Biología Celular y Molecular I. **Guest lecturer**. Universidad de Antioquia, Colombia
2018 Advanced Statistics for Ecologist using R. **Guest lecturer**. San Diego State University
2017 Climate Change Impact on Ecosystems. **Guest lecturer**. San Diego State University

Experimental Ecology. **Instructor**. San Diego State University
2016 Experimental Ecology. **Instructor**. San Diego State University
2013 Biostatistics. **Teaching Assistant**. San Diego State University
2010 Introduction to Wetlands. **Teaching Assistant**. University of Puerto Rico, Mayagüez
2009 Physical and Chemical Properties of Soils. **Teaching Assistant**. University of Puerto Rico, Mayagüez
2008 Physiological Principles of Crop Production. **Teaching Assistant**. University of Puerto Rico, Mayagüez
2007 Fundaments of Crop Production. **Teaching Assistant**. University of Puerto Rico, Mayagüez

TRAINING

Undergraduate students	**Graduate students**
G. Doney: 2025 – present	U.C. Morikwe 2026 – present
L. Barnett 2024	J. Falkowski 2024 – 2026
A. Coy 2024	L. Ordoñez 2017 – 2018